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Dublin City University

AUGUST/RESIT EXAMINATIONS 2022/2023

MODULE: CA208 - Logic

PROGRAMME(S):

COMSCI	BSc in Computer Science
CASE	BSc in Computer Applications (Sft.Eng.)
ECSAO	Study Abroad (Engineering & Computing).
ECSA	Study Abroad (Engineering & Computing).

YEAR OF STUDY: 2,O,X

EXAMINERS: Dr. David Sinclair (Internal) (Ext:5510)

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer ALL 6 questions from Section A. Answer 2 out of 3 questions from Section B. Question 1 carries 30 marks. Questions 2 to 9 carry 10 marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO

The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

Section A

QUESTION 1 ***[Total marks: 30]***

Heads you win. Tails I lose.

1(a) [10 Marks]

Convert the problem statement above into a set of propositional formulae.

Hint: The coin is either heads or tails but not both at the same time. Either you win or I win but not both at the same time. The following equivalence is useful: $A \oplus B \equiv \neg(A \rightarrow B) \vee \neg(B \rightarrow A)$

1(b) [10 Marks]

Convert the propositional formulae in Question 1(a) into *Conjunctive Normal Form* (CNF).

1(c) [10 Marks]

Using *Resolution*, prove you win.

[End Question 1]

QUESTION 2 ***[Total marks: 10]***

[10 Marks]

Without using truth tables, show that

$$(A \rightarrow C) \wedge (B \rightarrow C) \equiv (A \vee B) \rightarrow C$$

Clearly indicate which logical equivalance is used at each step.

[End Question 2]

QUESTION 3 ***[Total marks: 10]***

[10 Marks]

Using semantic tableaux, prove the following propositional formula is valid.

$$(p \rightarrow q) \rightarrow ((p \rightarrow \neg q) \rightarrow \neg p)$$

[End Question 3]

QUESTION 4 ***[Total marks: 10]***

[10 Marks]

Using the Gentzen System $\mathcal{G}$ prove

$$\vdash (A \to B) \to ((\neg A \to B) \to B)$$

[End Question 4]

QUESTION 5 ***[Total marks: 10]***

[10 Marks]

Using Gentzen system $\mathcal{G}$, prove the following predicate formula.

$$\forall x.(p(x) \to q(x)) \to (\forall x.p(x) \to \forall x.q(x))$$

[End Question 5]

QUESTION 6 ***[Total marks: 10]***

[10 Marks]

Convert the following formula to prenex conjunctive normal form (PCNF)

$$\exists x.\forall y.p(x,y) \to \forall y.\exists x.p(x,y)$$

[End Question 6]

Section B

QUESTION 7 ***[Total marks: 10]***

[10 Marks]

Let a set be represented by a list of numbers. Write the following Prolog relations:

- `union(A,B,C)` is true if the list `C` contains only even numbers in either lists `A` or `B`.
- `intersection(A,B,C)` is true if the list `C` contains only even numbers in both lists `A` and `B`.

[End Question 7]

QUESTION 8 ***[Total marks: 10]***

[10 Marks]

Write a Prolog relation `oddsquaresum(X, S)` that is true when `S` is the sum of the square of all the odd numbers in the list `X`. Note that the list `X` can contain any atom, e.g. `[4,oranges,5,2,7,apples,1]`. Hence, `oddsquaresum([4,oranges,5,2,7,apples,1], 75)` is true.

[End Question 8]

QUESTION 9 ***[Total marks: 10]***

[10 Marks]

Describe *cuts* in Prolog. Why are *cuts* used? Illustrate your answer with examples.

[End Question 9]

APPENDICES

Semantic Tableaux

α	α_1	α_2
$\neg\neg A_1$	A_1	
$A_1 \wedge A_2$	A_1	A_2
$\neg(A_1 \vee A_2)$	$\neg A_1$	$\neg A_2$
$\neg(A_1 \rightarrow A_2)$	A_1	$\neg A_2$
$\neg(A_1 \uparrow A_2)$	A_1	A_2
$A_1 \downarrow A_2$	$\neg A_1$	$\neg A_2$
$A_1 \leftrightarrow A_2$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$
$\neg(A_1 \oplus A_2)$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$

β	β_1	β_2
$\neg(B_1 \wedge B_2)$	$\neg B_1$	$\neg B_2$
$B_1 \vee B_2$	B_1	B_2
$B_1 \rightarrow B_2$	$\neg B_1$	B_2
$B_1 \uparrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(B_1 \downarrow B_2)$	B_1	B_2
$\neg(B_1 \leftrightarrow B_2)$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$B_1 \oplus B_2$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$

γ	$\gamma(a)$
$\forall x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$

δ	$\delta(a)$
$\exists x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$

Gentzen Logic

$$\frac{\vdash U_1' \cup \{\alpha_1, \alpha_2\}}{\vdash U_1' \cup \{\alpha\}} \qquad \frac{\vdash U_1' \cup \{\beta_1\} \qquad \vdash U_2' \cup \{\beta_2\}}{\vdash U_1' \cup U_2' \cup \{\beta\}}$$

$$\frac{\vdash U_1' \cup \{\gamma, \gamma(a)\}}{\vdash U_1' \cup \{\gamma\}} \qquad \frac{\vdash U_1' \cup \{\delta(a)\}}{\vdash U_1' \cup \{\delta\}}$$

$\frac{A}{B}$ is a *sequent* and means that if A is true, then B can be inferred.

α	α_1	α_2
$\neg\neg A$	A	
$\neg(A_1 \wedge A_2)$	$\neg A_1$	$\neg A_2$
$A_1 \vee A_2$	A_1	A_2
$A_1 \rightarrow A_2$	$\neg A_1$	A_2
$A_1 \uparrow A_2$	$\neg A_1$	$\neg A_2$
$\neg(A_1 \downarrow A_2)$	A_1	A_2
$\neg(A_1 \leftrightarrow A_2)$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$
$A_1 \oplus A_2$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$

β	β_1	β_2
$B_1 \wedge B_2$	B_1	B_2
$\neg(B_1 \vee B_2)$	$\neg B_1$	$\neg B_2$
$\neg(B_1 \rightarrow B_2)$	B_1	$\neg B_2$
$\neg(B_1 \uparrow B_2)$	B_1	B_2
$B_1 \downarrow B_2$	$\neg B_1$	$\neg B_2$
$B_1 \leftrightarrow B_2$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$\neg(B_1 \oplus B_2)$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$

γ	$\gamma(a)$
$\exists x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$

δ	$\delta(a)$
$\forall x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$

[END OF APPENDICES]

[END OF EXAM]